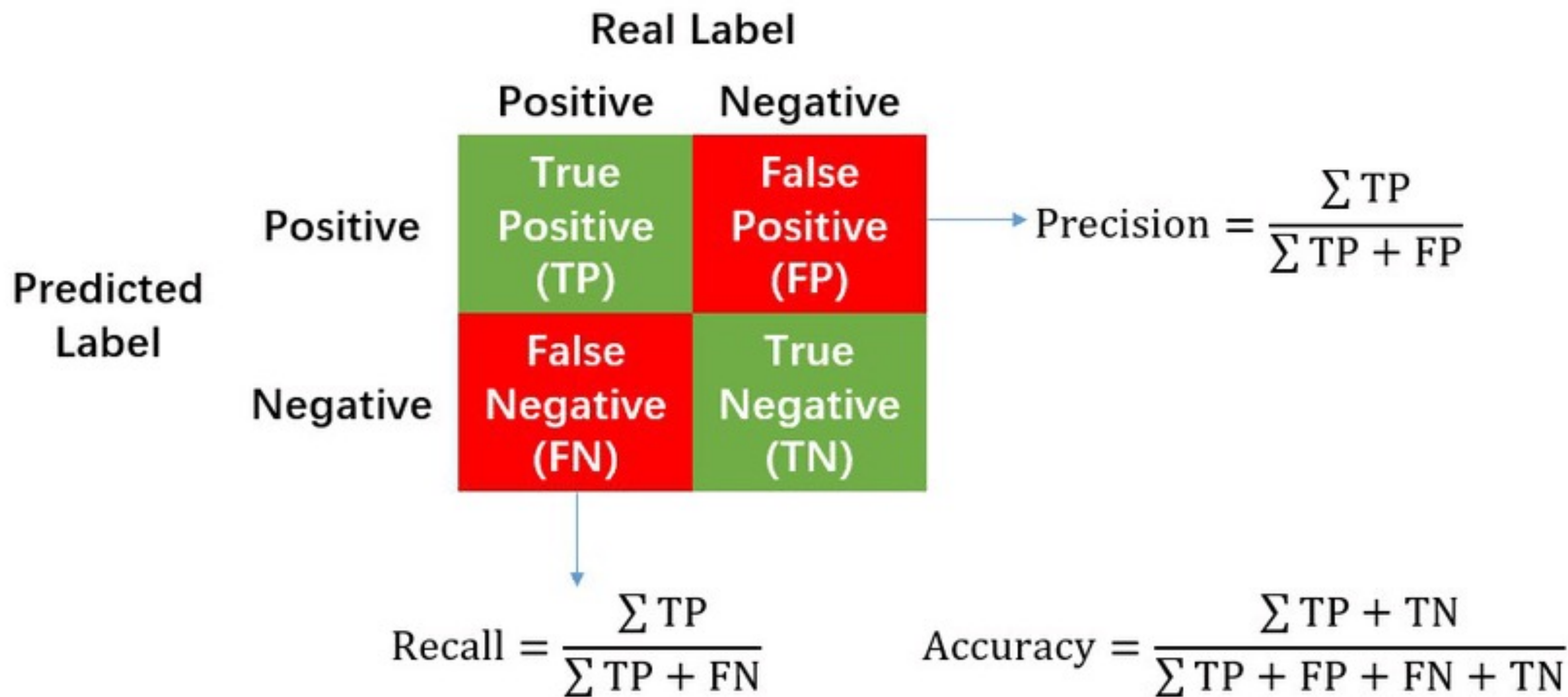


LLM Evaluation



Precision/Recall

- When computing one of the two measures, it is important (if possible) to report the other
- The reason is that it is easy to increase one at the expense of the other
 - Predicting more positive would increase recall at the expense of precision
 - Predicting more negative would increase precision at the expense of recall
- Often their harmonic mean is used to report one is known as F-score:

$$\text{F-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Language Model Evaluation

- There are no good ways to evaluate language models
- Common practices focus on precision
- Recall is hard because
 - language models are used for very diverse goals – goals are often vague or implicit
 - there are generally many very different ways to correctly follow an instruction or a set of examples – difficult to enumerate them
 - Difficult also for (trained) humans

Language Model Evaluation

- Precision is somewhat simpler
 - There are increasingly better automatic metrics for determining when two texts are equivalent (albeit, far from perfect)
 - Precision annotation usually much less cognitively taxing for humans than recall
- Several directions are employed:
 - Multiple-choice tests / closed set of answers
 - Using embedding-based metrics for comparing between machine output and a reference
 - Prompting the models for judgements (but – may be biased)

Language Model Evaluation

- Several directions are employed:
 - Multiple-choice tests

Microeconomics	One of the reasons that the government discourages and regulates monopolies is that	
	(A) producer surplus is lost and consumer surplus is gained.	✗
	(B) monopoly prices ensure productive efficiency but cost society allocative efficiency.	✗
	(C) monopoly firms do not engage in significant research and development.	✗
	(D) consumer surplus is lost with higher prices and lower levels of output.	✓

Figure 3: Examples from the Microeconomics task.

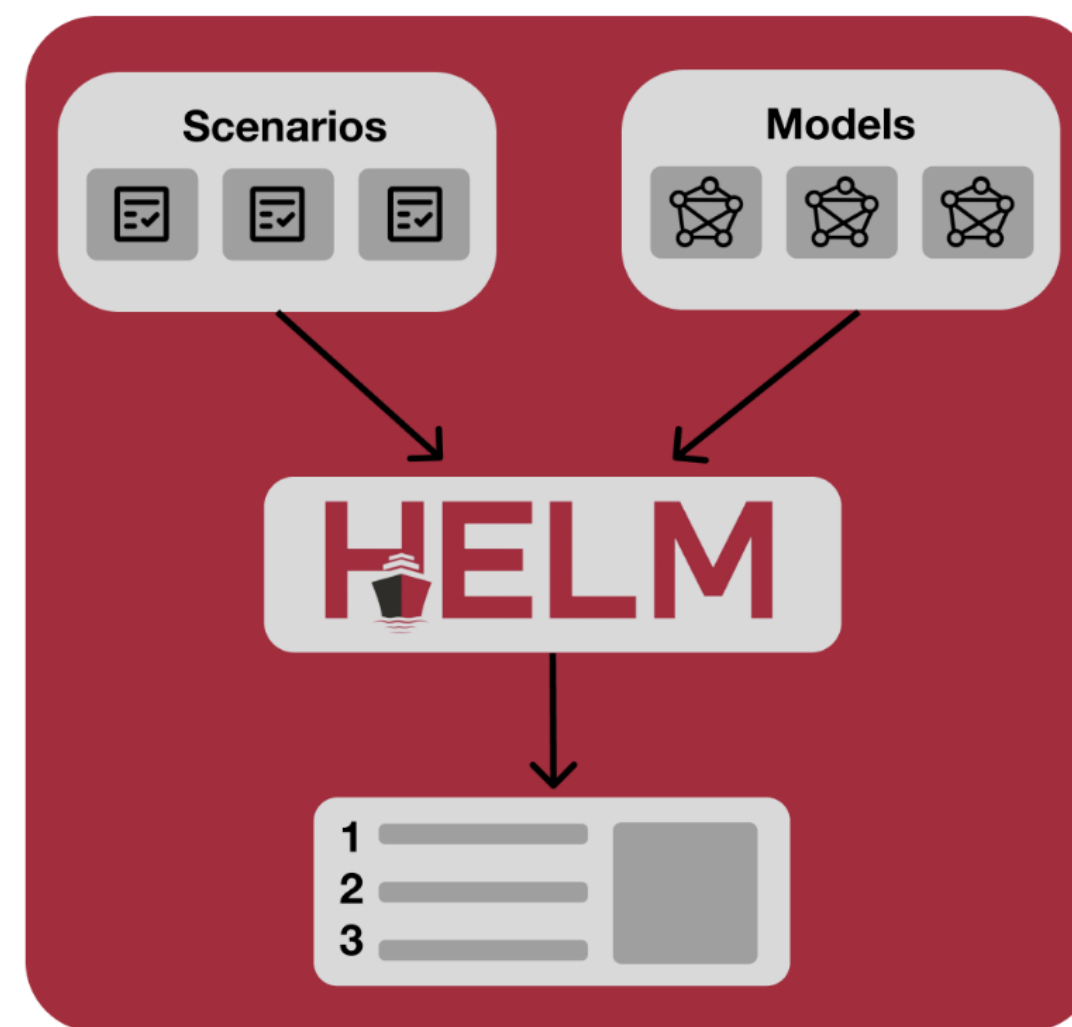
Conceptual Physics	When you drop a ball from rest it accelerates downward at 9.8 m/s^2 . If you instead throw it downward assuming no air resistance its acceleration immediately after leaving your hand is	
	(A) 9.8 m/s^2	✓
	(B) more than 9.8 m/s^2	✗
	(C) less than 9.8 m/s^2	✗
	(D) Cannot say unless the speed of throw is given.	✗
College Mathematics	In the complex z -plane, the set of points satisfying the equation $z^2 = z ^2$ is a	
	(A) pair of points	✗
	(B) circle	✗
	(C) half-line	✗
	(D) line	✓

Figure 4: Examples from the Conceptual Physics and College Mathematics STEM tasks.

Language Model Evaluation

- Benchmarks of many tests
 - HELM
 - SuperGLUE

A holistic framework for evaluating foundation models.



Model	Mean win rate
GPT-4 (0613)	0.965
GPT-4 Turbo (1106 preview)	0.842
Palmyra X V3 (72B)	0.832
Palmyra X V2 (33B)	0.794
PaLM-2 (Unicorn)	0.784
Yi (34B)	0.781

SEE MORE

Language Model Evaluation

- Embedding-based reference-based metrics

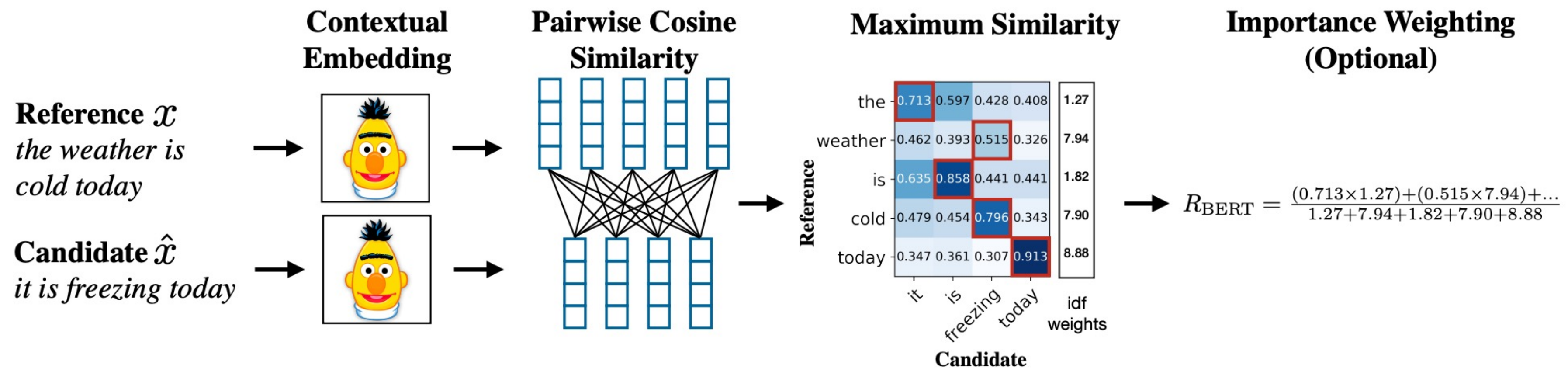
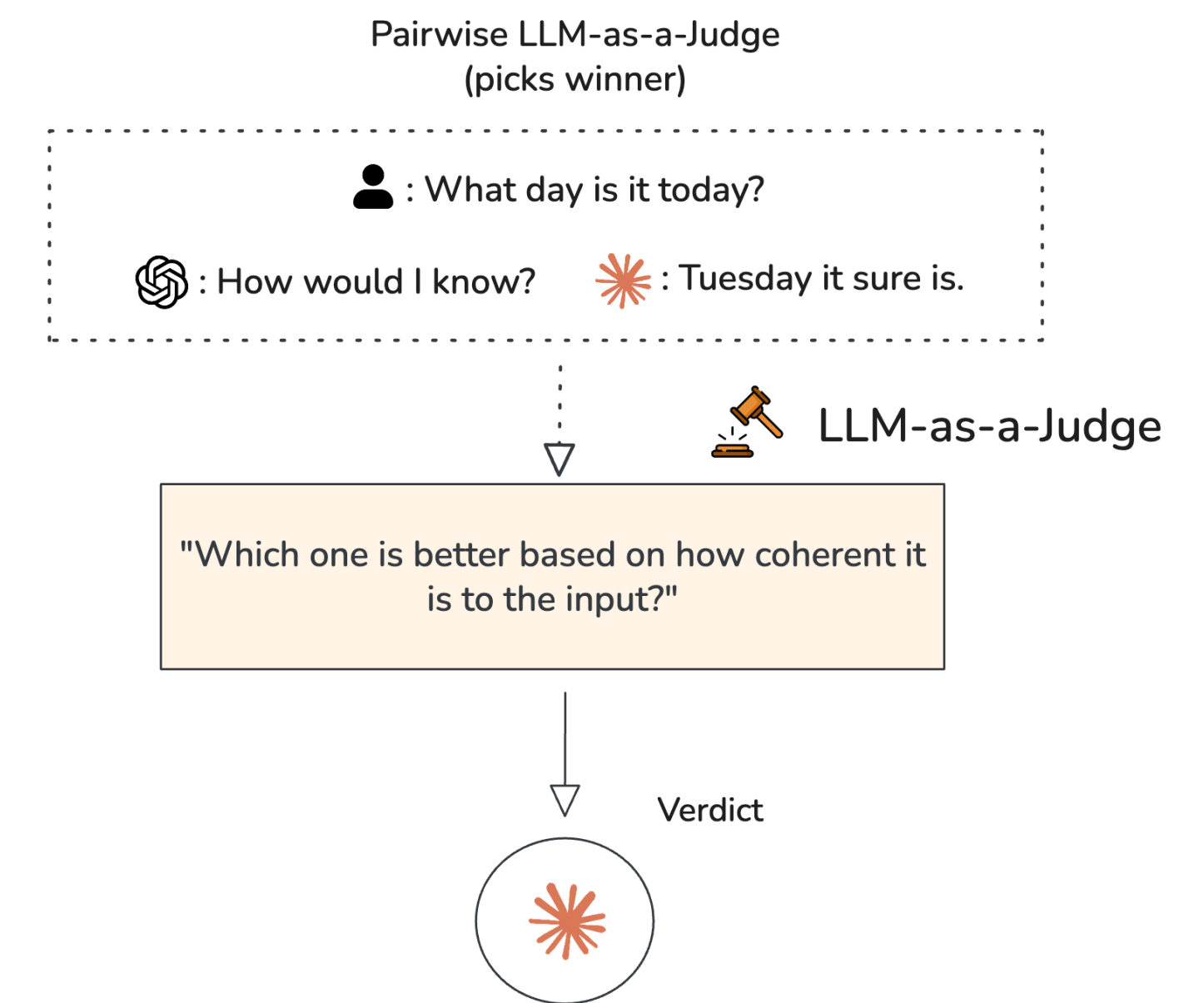
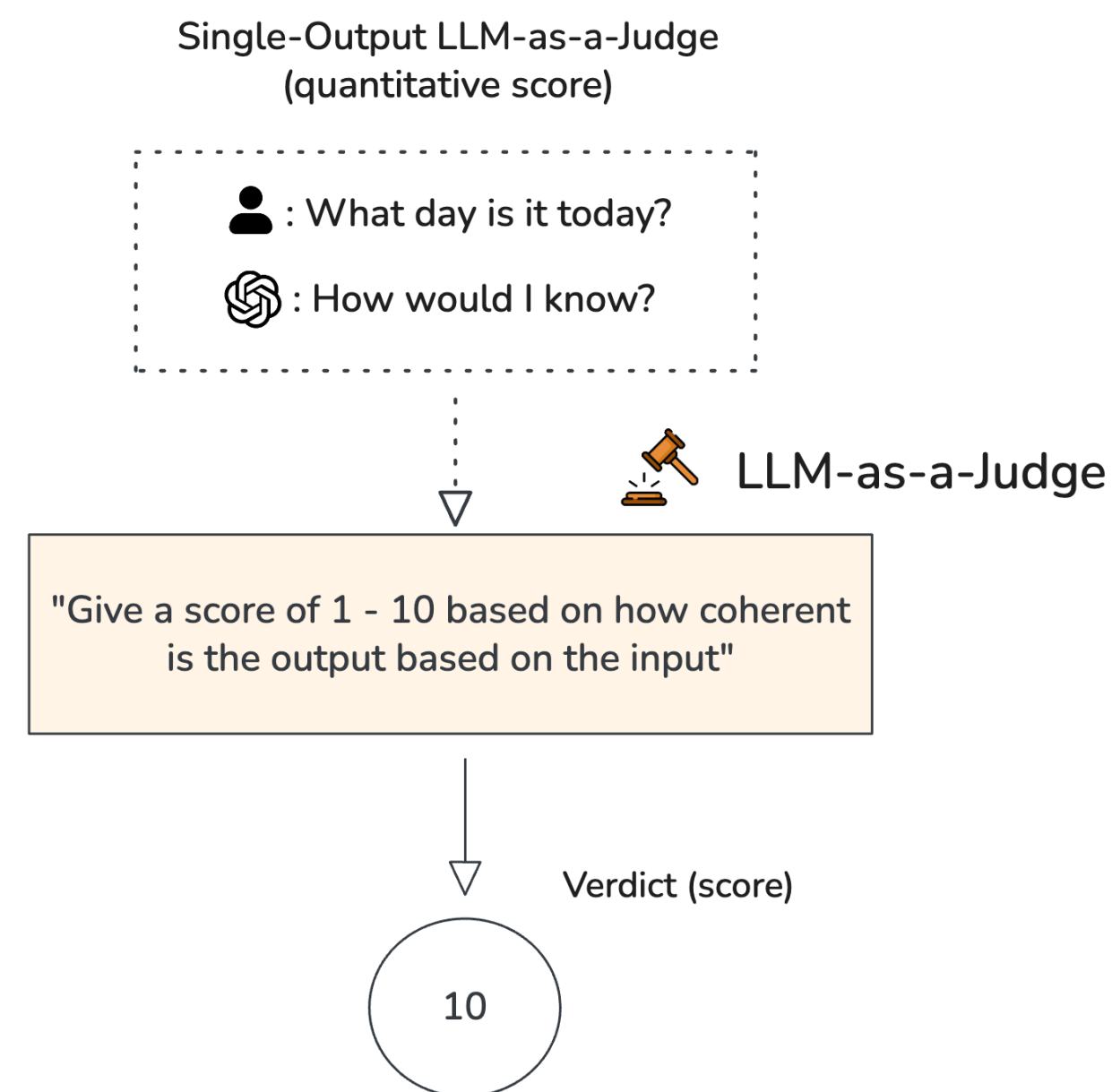


Figure 1: Illustration of the computation of the recall metric R_{BERT} . Given the reference x and candidate $\hat{x}$, we compute BERT embeddings and pairwise cosine similarity. We highlight the greedy matching in red, and include the optional idf importance weighting.

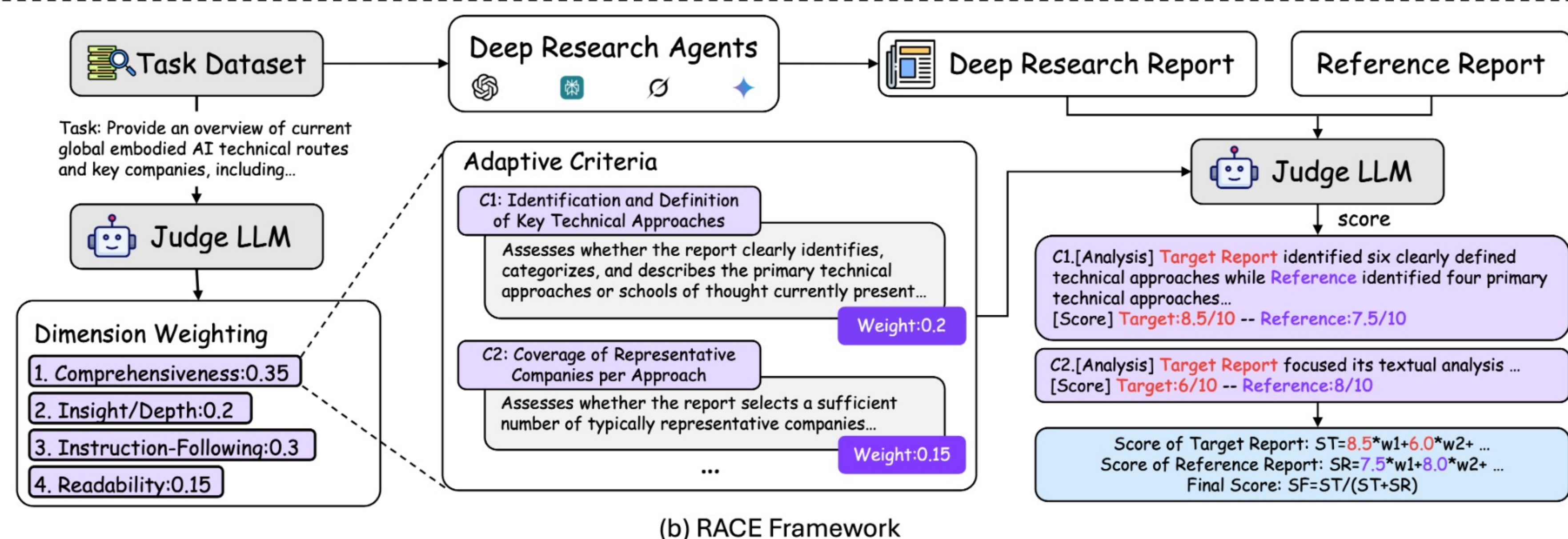
Language Model as Judge (LMJ) Methodology

- Let an LLM judge the output of another LLM (usually a weaker one)
- This is similar to the approach on the previous slide, but uses no reference
- Two main varieties:
 - Pair-wise
 - Single output (grading)
 - Multiple outputs



Language Model as Judge (LMJ) Methodology

- Multiple outputs are usually needed where the answer has several dimensions
 - Often the exact dimensions are undisclosed to avoid over-fitting



LMJ Methodology

- Problems include
 - circular definition of the task
 - biases of the LMJ may be in the same cases where LMs fail to produce correct output
 - lack of standardized validation the methodology (but see Calderon et al., 2025)